

Introduction

This SGTA introduces you to a study which will be used for the rest of STAT6190 (`health.csv`, available on iLearn). This study investigates the effects of different diet and exercise interventions on a variety of physiological, behavioural, and self-reported health outcomes in a diverse adult sample. A total of 1,500 participants were recruited from four urban gyms and randomly assigned to combinations of diet plans (Low Carb, Mediterranean, Intermittent Fasting) and exercise regimens (Cardio, Strength, Mixed). Data collected include anthropometric measures (e.g., weight loss), cardiovascular indicators (e.g., resting heart rate), metabolic markers (e.g., cholesterol), psychological and behavioural measures (e.g., reaction time, satisfaction). The aim is to assess the effects of interventions, test group differences, and evaluate key assumptions of classical statistical methods in a realistic setting.

The variables in the dataset are listed below:

- **id**: participant identifier
- **gender**: Male/Female. Categorical
- **age**: age in years. Integer
- **gym**: one of East, South, West, North. Categorical
- **diet**: one of Mediterranean, Intermittent Fasting, Low Carb. Categorical
- **exercise**: one of Cardio, Mixed, Strength. Categorical
- **fitness_score_before**: fitness score (0-100) at enrolment in the study. Integer
- **fitness_score_after**: fitness score (0-100) at the end of the study. Integer
- **weight_loss**: weight loss in kg. Numerical continuous.
- **cholesterol_before**: cholesterol level at enrolment in the study. Integer
- **cholesterol_after**: cholesterol level at the end of the study. Integer
- **met_goal**: if the goal (5kg weight loss) has been met. 0/1
- **success**: if the fitness score has improved by 5. 0/1
- **satisfaction**: the level of satisfaction (1-5). Categorical ordinal
- **week1**: the weight loss in week 1. Numerical continuous
- **week2**: the weight loss in week 2. Numerical continuous
- **week3**: the weight loss in week 3. Numerical continuous
- **week4**: the weight loss in week 4. Numerical continuous
- **rhr_before**: Resting Heart Rate (RHR) at enrolment in the study. Numerical continuous
- **rhr_after**: Resting Heart Rate (RHR) at the end of the study. Numerical continuous
- **reaction_time_before**: Reaction time (RT) in milliseconds at enrolment in the study. Numerical continuous
- **reaction_time_after**: Reaction time (RT) in milliseconds at the end of the study. Numerical continuous

Write a report in RMarkdown and address the following task.

1. EDA (Exploratory Data Analysis). Use numerical and graphical tools to illustrate the data, including single-variable summaries and relations among variables. This week, focus your efforts on the variables considered in the questions below.

2. Answer the following questions using appropriate tests, which you can identify from the questions themselves and checking if the test assumptions are satisfied. When the assumptions are not satisfied, try different strategies like **stratification by gender** or variable transformation.
- Is the fitness score before the study equal to 50?
 - Is the mean weight loss different from 5 kg?
 - Did participants improve their fitness scores after 8 weeks?
 - Is the mean/median change in cholesterol significantly different from 0?
 - Is the reaction time at the enrolment in the study equal to 275ms? *Can we answer this question or do we need to stratify by gender? If we can answer the question, do it for the whole cohort, otherwise explain why we need to stratify by gender and answer the research question for both males and females.*
 - Has the Reaction time improved at the end of the study? *Can we answer this question or do we need to stratify by gender? If we can answer the question, do it for the whole cohort, otherwise explain why we need to stratify by gender and answer the research question for both males and females.*